Library Management System

Library Management System

Here's the conclusion in our Library Management System analysis:

- Some categories have a higher proportion of missing books, indicating a need for stronger monitoring and security.
- Certain books are checked out multiple times in a single day, showing strong demand and user preference.
- Most popular categories and authors were identified, useful for planning future acquisitions.
- Top 3 authors contribute significantly to the total checkouts.
- Some authors' books are missing more frequently, requiring closer attention.
- Cabinet and rack/row analysis revealed:
    - Weak signal strength in certain cabinets.
    - Specific rack/row combinations are prone to missing books.
    - One cabinet has the highest proportion of checkouts.
    - Some cabinet/rack/row spaces remain unused.
- Peak times of book checkouts were identified, showing user activity trends.

Key Recommendations

- Strengthen Book Monitoring
    - Implement stricter tracking for categories/authors with high missing rates.
    - Introduce RFID/barcode scanning for better real-time inventory management.
- Optimize Storage Utilization
    - Reorganize cabinets/racks to reduce overcrowding and fill unused spaces.
    - Improve signal strength in weak cabinets to enhance digital tracking.
- Enhance Resource Allocation
    - Increase copies of books/authors that are most frequently checked out.
    - Expand collections in the most popular categories.
- User Engagement & Service Improvement
    - Use peak checkout time data to schedule staff for better service.
    - Recommend high-demand books to users through alerts or notices.
- Loss Prevention
    - Focus on rack/row combinations prone to missing books.
    - Set up monitoring or CCTV in areas with repeated losses.

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Books that have been checked out multiple times in a single day

```sql
-- books that have been checked out multiple times in a single day
select Title,count(*) as Total_CheckOut,Date(Timestamp)
 from lib where status = "Checked Out"
 group by Title , date(Timestamp)
 order by Total_CheckOut desc  limit 1
```

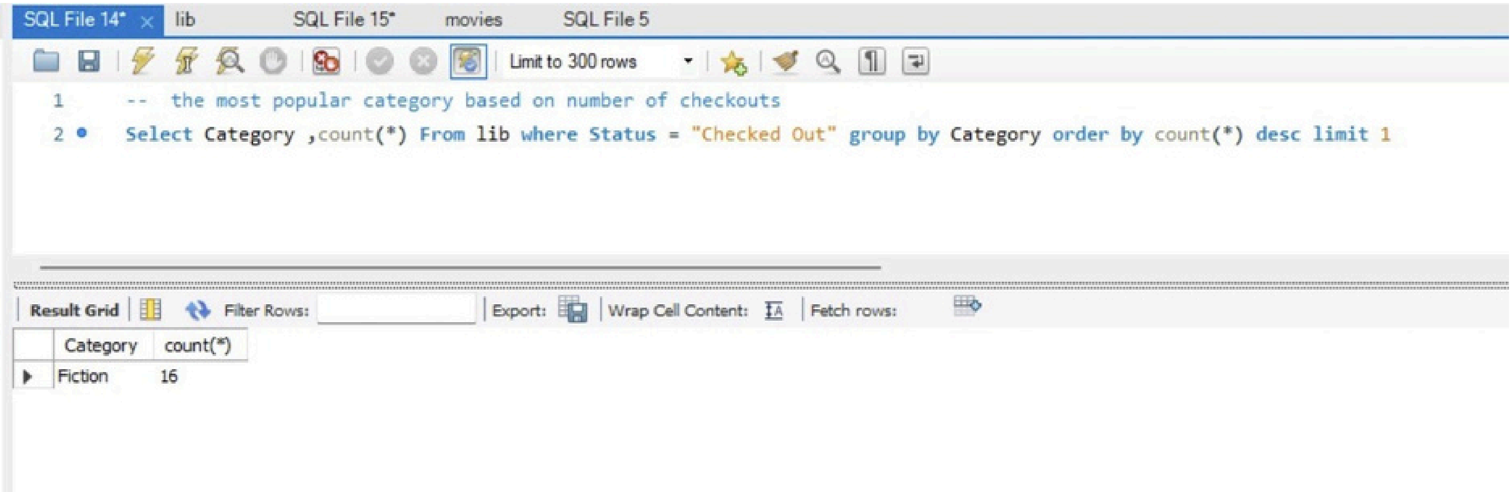
Title	Total_CheckOut	Date(Timestamp)
The Quest of Future	4	2024-12-11

Category has the highest of missing books compared to total books in that category

```sql
select  Category,count(*) as Count from lib where Status = "Missing"
 group by Category order by Count desc limit 1;
```

Category	Count
Programming	16

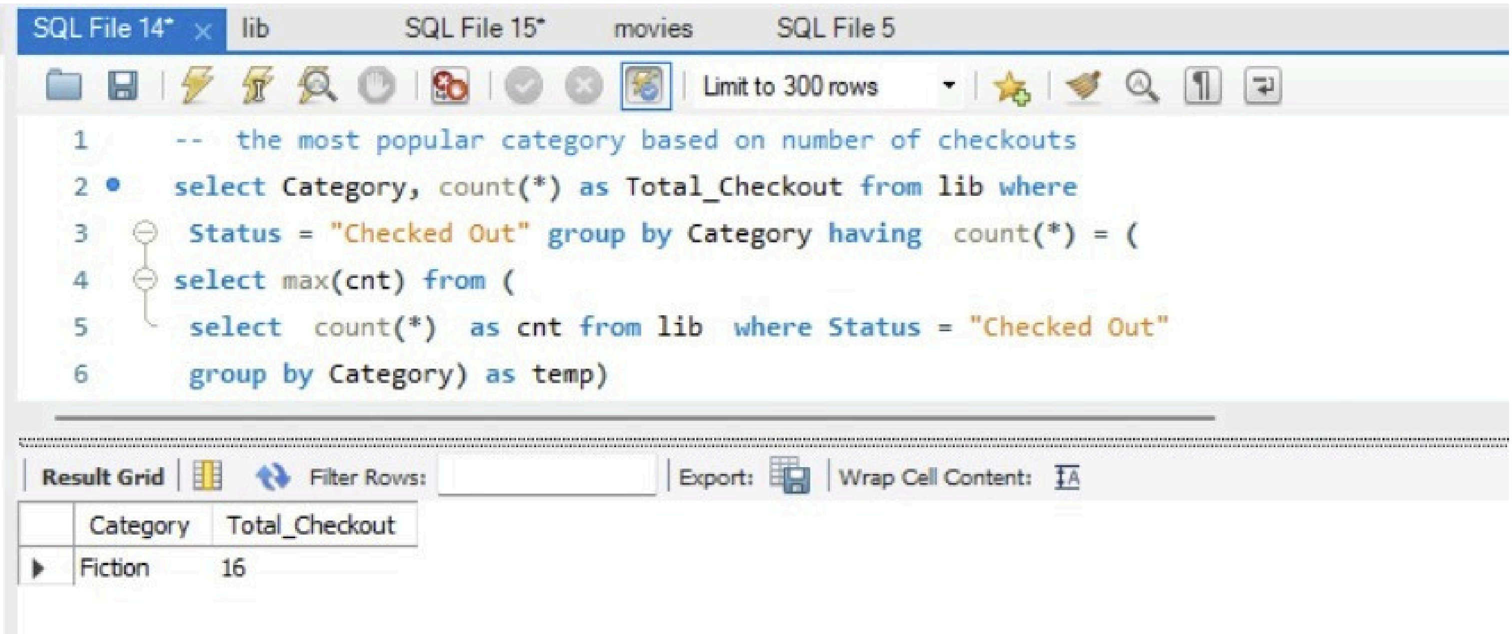
The most popular category based on number of checkouts




```sql
-- the most popular category based on number of checkouts
Select Category ,count(*) From lib where Status = "Checked Out" group by Category order by count(*) desc limit 1
```

Category	count(*)
Fiction	16

The most popular category based on number of checkouts (through sub- query)




```sql
-- the most popular category based on number of checkouts
select Category, count(*) as Total_Checkout from lib where
 Status = "Checked Out" group by Category having  count(*) = (
select max(cnt) from (
 select  count(*)  as cnt from lib  where Status = "Checked Out"
 group by Category) as temp)
```

Category	Total_Checkout
Fiction	16

The author writes across the maximum number of categories

```sql
-- author writes across the maximum number of categories
select  author, count(*)  as  "Total_Book_Written_Across_Category"
  from lib Group by Author, Category
Order by Total_Book_Written_Across_Category
desc limit 1;
```

author	Total_Book_Written_Across_Category
Taylor Anderson	4

Cabinet has the weakest average signal strength

```sql

-- Cabinet has the weakest average signal strength
select Cabinet from lib  where Signal_Strength = (
                                  select min(Signal_Strength) from lib)
```

Cabinet
2
2
3
4

Rack/Row combination where books are most frequently missing

```sql
-- Rack/Row combination where books are most frequently missing
select Rack,lib.row  from lib  where
Status ="Missing" group by Rack , lib.Row having count(*) = (
select max(cnt) from(
select  count(*) cnt from lib  where
Status ="Missing" group by Rack , lib.Row
) as temp)
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Rack	row
2	3

The top 3 authors with the most checked-out books

```sql
-- the top 3 authors with the most checked-out books
select Author,count(*) as Total_Count
from lib group by Author having count(*) = (
select max(cnt) from (select count(*) as cnt from lib where Status  = "Checked Out"  group by Author) as temp
)
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Author	Total_Count
Morgan Clark	3
Dakota Smith	3
Taylor Taylor	3
Riley Lee	3
Jordan Lee	3
Alex Anderson	3
Skyler Brown	3
Alex Lee	3
Riley Allen	3
Alex Brown	3

The Author’s Books are Most Frequently Missing

SQL File 14* | lib | SQL File 15* | movies | SQL File 5

Limit to 300 rows

```sql
-- author's books are most frequently missing
select Author, count(*) Missing_Values
 from lib where Status = "Missing"
 group by Author  order by Missing_Values
                                desc  limit 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Author	Missing_Values
Taylor Brown	5

The Cabinet that has the highest proportion of checked-out books

SQL File 14* | lib | SQL File 15* | movies | SQL File 5 | lib

Limit to 300 rows

```sql
--  the Cabinet that has the highest proportion of checked-out books
select Cabinet  from lib where Status ="Checked Out"  group by Cabinet having count(*) = (select max(cnt)from
                                     (select Count(*)  as cnt
                                             from lib where Status ="Checked Out"  group by Cabinet) as temp )
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Cabinet
2

Unused spaces (Cabinet/Rack/Row) where no books are present

SQL File 14* | lib | SQL File 15* | movies | SQL File 5 | lib

Limit to 300 rows

```sql
-- Unused spaces (Cabinet/Rack/Row) where no books are present
select Cabinet,Rack, lib.Row  from lib where
Status In ("Checked Out","Missing")
group by Cabinet,Rack, lib.Row
order by Cabinet,Rack, lib.Row
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Cabinet	Rack	Row
1	1	2
1	1	3
1	1	4
1	1	5
1	2	3
1	2	4
1	3	1
1	3	2
1	3	4
1	4	1
1	4	2
1	4	3
1	5	1

The peak time of day when books are checked out most

SQL File 14* | lib | SQL File 15* | movies | SQL File 5 | lib

Limit to 300 rows

```sql
-- the peak time of day when books are checked out most
select Timestamp , count(*) as most_frequent_timestamp
from  lib  group by Timestamp having  count(*)= (
(select max(cnt) from (
select count(*) cnt  from lib  group by Timestamp) as temp )) limit 1
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Timestamp	most_frequent_timestamp
2024-12-11 10:01:00	1